PhD Lecture in Behavioral Asset Pricing: Bounded Rationality and Psychology-free Models

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Overview

- **Bounded rationality:** people get only complicated or subtle things wrong
  - Broad theme: we have limited capacity for gathering and processing information
- **Specific topics:**
  - **Limited attention**
    - Underreaction to news
    - Buying vs. selling decisions of individual investors
    - Causal effect of attention on asset prices
  - **Category thinking**
- **Psychology-free models:** we can make interesting predictions without explicit use of psychological concepts
  - Actions of sophisticated traders
  - Actions of noise traders
  - Impacts of investor sentiment (equity and credit market sentiment)

Limited Attention

Limited Attention I

- **Theme:**
  - Because of limited attention, investors may underreact to news
- **Underreaction to earnings news**
  - Limited attention predicts weaker immediate market reaction and stronger post-earnings announcement drift (PEAD)
- **Two influential studies:**
  - PEAD is stronger for firms that announce earnings at the same time as many other firms (Hirshleifer, Lim, and Teoh, 2009)
  - PEAD is stronger for firms that announce earnings on Friday (Della Vigna and Pollet, 2009)
  - These two studies help rule out the alternative explanation that the differential PEAD is due to differential information content of earnings surprises

Limited Attention I, ctd.

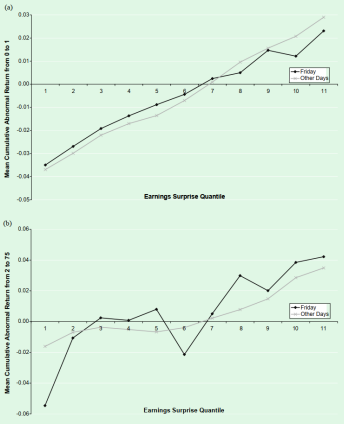
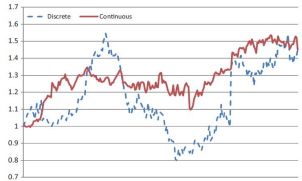


Figure 1a and b. Immediate response to earnings surprises and delayed response to earnings surprises. Stocks in CRSP are matched to quarterly earnings announcements in I/B/E/S from January 1995 until June 2006. In event time, day 0 is the day of the announcement. The cumulative abnormal return for each stock is the raw buy-and-hold return adjusted using the estimated beta from market model. The earnings surprise for an announcement is the difference between actual earnings for the quarter recorded by I/B/E/S and the median analyst forecast included in the I/B/E/S detail file during the 30 days before the quarterly earnings announcement scaled by the stock price 5 trading days before the announcement. Quantiles 1 to 5 contain earnings announcements for five quintiles of negative earnings surprises and quantiles 7 to 11 contain earnings surprises for five quintiles of positive earnings surprises. Quantile 6 contains all announcements with an earnings surprise equal to zero. The breakpoints for the quantiles are determined separately for each year. Announcements made on Saturday or Sunday are excluded from the sample.

Della Vigna and Pollet (2009): Friday earnings announcements exhibit a *weaker immediate market reaction* but a *stronger post-earnings announcement drift (PEAD)* than announcements on other days.

Limited Attention I, ctd.

- **A potential explanation for momentum** (Da, Gurun, and Warachka, 2014)
  - A series of frequent small changes attracts less attention than infrequent large changes
  - "Frog in the Pan" hypothesis: investors are more likely to be inattentive to information arriving continuously in small amounts than to information that arrives in large amounts at discrete time points
  - Momentum is stronger when information arrives continuously as opposed to discretely



Frog in the pan: small, continuous, overlooked

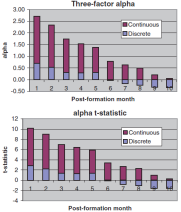


Figure 2
Momentum profits
This figure plots risk-adjusted momentum profits in the continuous and discrete information portfolios from one to ten months after portfolio formation during the post-1927 sample period. ID is defined in Equation (1) to capture the distribution of daily returns across the formation period. Continuous information arrives frequently in small amounts, while discrete information arrives infrequently in large amounts. Momentum profits in month *t* + 1, where *t* ranges from 1 to 10, based on double-sorted portfolios formed in month *t* according to formation-period returns and ID. These momentum profits are not cumulative. Instead, they are time-series averages of holding-period returns at a single month after portfolio formation, with the month of portfolio formation varying across the sample period.

Limited Attention I, ctd.

- Underreaction is more likely to occur when the news is **less salient or more subtle**
- **Underreaction to demographic information**
  - Della Vigna and Pollet (2008): investors are slow to recognize the impact of demographic shifts on the future profitability of firms with age-sensitive products
  - E.g. Baby boom today predicts higher demand for toys in 6 years, so prices of toy-making firms should rise immediately in an efficient market
- **Underreaction to technological information**
  - Innovative efficiency: Hirshleifer, Hsu, and Li (2013)
  - Track records in innovation ability: Cohen, Diether, and Malloy (2013)
- **Underreaction to the absence of news: No news is news**
  - Giglio and Shue (2014): investors underreact to the probability of merger deal completion contained in the passage of time

My Research: The Information Content of Sudden Insider Silence

- **Hong and Li (JFQA 2019)**, "The Information Content of Sudden Insider Silence"
  - Identifies "routine" insiders from their calendar trading patterns
  - Sudden silence is a deviation from routine trading patterns
  - Routine insiders strategically stay silent when they hold private information
  - Silence after routine selling (buying) predicts positive (negative) future returns and fundamentals
  - A long-short strategy earns 6–10% annually
  - Returns concentrate at earnings announcements and are stronger where information environments are poor and arbitrage is costly
- **Connection:** the previous slide shows investors underreact to subtle news; here they underreact to the *absence* of an expected event, because it is hard to recognize that nonoccurrence can be informative

Limited Attention I, ctd.

- A growing literature documents that investors underreact to news of economically linked peers ("momentum spillover effect")
- **Customer-Supplier Links** (Cohen and Frazzini, 2008)
  - Firms are required to report their major customers (>10% of sales)
  - Investors are slow to recognize that good news for a customer is good news for the supplier

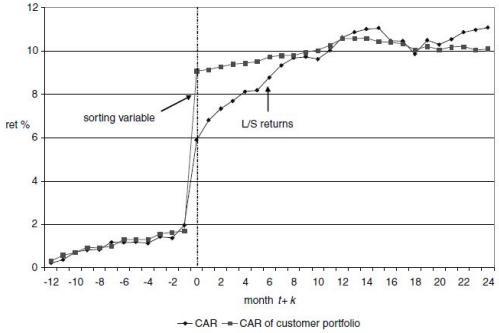


Figure 2. Customer momentum, event-time CAR. This figure shows the average cumulative return in month $t+k$ on a long–short portfolio formed on the firm's customer return in month t. At the beginning of every calendar month, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous month. Stocks are assigned to one of five quintile portfolios. The figure shows average cumulative returns (in %) over time of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer returns stocks.

Cohen and Frazzini (2008)

Limited Attention I, ctd.

- The same pattern holds for other types of economic linkages
- **Industry and product market peers**
  - Moskowitz and Grinblatt (1999): firms operating in the same industry (industry momentum)
  - Cohen and Lou (2012): conglomerate firms are slower than standalone firms to reflect industry news
  - Huang (2015): stock prices of multinational firms underreact to foreign industry news
  - Hoberg and Phillips (2016): product market peers identified through annual reports
- **Geographic peers**
  - Parsons, Sabbatucci, and Titman (2020): firms headquartered in the same state
  - Jin and Li (2024): firms with overlapping geographic establishments
- **Technological peers**
  - Lee et al. (2018): firms sharing the same type of technologies
- **Analyst co-coverage**
  - Ali and Hirshleifer (2019): firms followed by the same set of analysts

My Research: Geographic and Scope Links

- **Jin and Li (JBFA 2024), geographic links**
  - Links firms through overlapping establishment locations (where sales are generated), not headquarters
  - Returns of geography-linked peers predict focal-firm returns and fundamentals
  - The effect is stronger for low-attention firms, costly-to-arbitrage stocks, and during high-sentiment periods
- **Jin and Li (working paper), scope similarity**
  - Links firms through Hoberg–Phillips product-market scope that transcends industry boundaries; investors are slow to incorporate news about scope peers, and a long-short portfolio earns 8.9% per year in five-factor alpha
- **Connection:** the momentum-spillover evidence on the previous slides relies on named linkages (customers, industries); here the linkage itself is the subtle information investors fail to process

Limited Attention I, ctd.

- Several papers document significant return predictability across debt and equity securities of the same firm
- Han, Subrahmanyam, and Zhou (JFE 2017)
  - The term structure of firms' credit spreads predicts future firm fundamentals and stock returns
- Addoum and Murfin (RFS 2020)
  - Prices of syndicated loans contain useful information for equity prices but investors fail to unravel such information, leading to return predictability from the loan market to the equity market

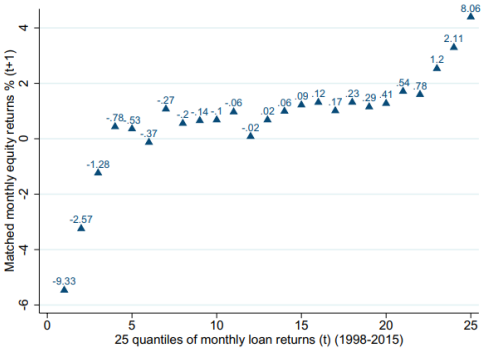


Figure 1
Equity returns versus lagged loan returns
The above plot combines both time-series and cross-sectional relationships between loan returns and linked equity returns in the next month. Loan returns are sorted into twenty-five quantiles over the entire sample period and plotted against the next month's equity returns for the same borrowers. Average loan returns for each quantile are reported next to the triangles.

Addoum and Murfin (RFS 2020)

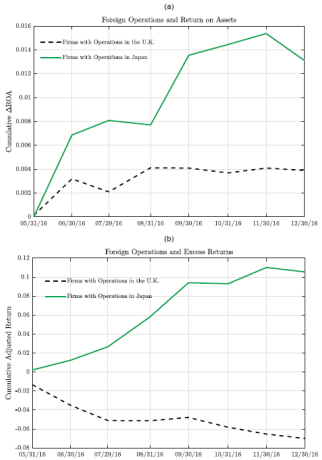
Limited Attention I, ctd.

- The source of underreaction could be "market segmentation"
  - Both debt and equity are claims to the same underlying cash flows/asset values
  - Investors trading in debt markets have low overlap with those trading in equity markets, leading to underreaction to cross-asset information
  - Supporting this explanation, the returns to cross-asset momentum are lower among equities held by mutual funds also trading in debt securities (dual ownership)

Limited Attention I, ctd.

- Firm value is also affected by shocks to other asset classes, including currencies and commodities
- **Cross-asset return predictability**
- Bae, Da, and Zurita (2023)
  - Equity investors of U.S. multinationals underreact to firms' currency shocks from the countries in which they operate
- Ho and Lauwers (2023)
  - Aggregate positions of traders in the commodity futures market can predict the cross-section of stock returns for commodity producers

Figure 1. (Color online) Foreign Currencies and Operations: The Brexit Case



Ho and Lauwers (2023)

Limited Attention II

- **Theme:**
  - Attention is more important for individual investors' buying decisions than for their selling decisions (Why?)
  - Barber and Odean (2008) show that there is indeed stronger buying interest than selling interest, among individual investors, for attention-grabbing stocks
  - Measure attention with extreme returns, high volume, or news announcements

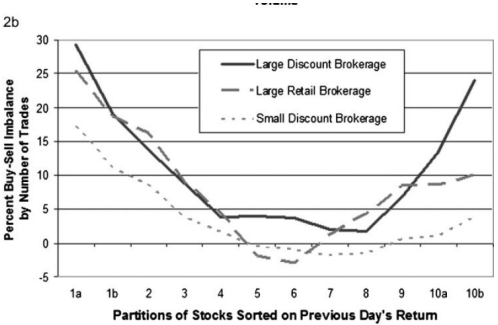


Figure 2
Individual Investor Buy-Sell Imbalances by Number of Trades for Stocks Sorted on the Current Day's Abnormal Trading Volume and Previous Day's Return

Limited Attention III: Hong and Stein (1999)

One primitive shock: news about future fundamentals diffuses gradually across investors.

1. Newswatchers: gradual information diffusion

- Population split into z groups; each period another group observes a subinnovation of

$$D_T = D_0 + \sum_{\tau=1}^{T} \varepsilon_\tau, \qquad \varepsilon_\tau = \varepsilon_{\tau,1} + \cdots + \varepsilon_{\tau,z}.$$

- CARA traders use private signals, but not current or past prices; they trade as if buy-and-hold.
- Hence price incorporates news linearly over z periods:

$$P_t = D_t + \frac{(z-1)\varepsilon_{t+1} + (z-2)\varepsilon_{t+2} + \cdots + \varepsilon_{t+z-1}}{z} - \theta Q.$$

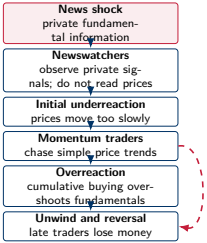
giving underreaction: positive short-run autocorrelation, no overshoot.

2. Momentum traders: simple trend-chasing

- Each generation holds for j periods and uses only a univariate signal:

$$F_t = A + \phi \Delta P_{t-1}, \qquad \phi > 0.$$

- They cannot tell whether ΔP_{t-1} reflects fresh news or past momentum trades.



Core mechanism Momentum trading accelerates the initial adjustment, but simple strategies convert underreaction into overshooting and reversals. The dashed arrow is the externality: early momentum traders induce late traders to buy.

Classroom hook: underreaction is not eliminated by arbitrage; it generates the profit opportunity that draws in momentum traders.

Source: Hong and Stein (1999), *Journal of Finance* 54(6), 2143–2184

Limited Attention III: Predictions and Comparative Statics

Equilibrium fixed point

Momentum-trader aggressiveness ϕ solves

$$\phi = \frac{\gamma \operatorname{cov}(P_{t+j} - P_t, \Delta P_{t-1})}{\operatorname{var}(\Delta P) \operatorname{var}_M(P_{t+j} - P_t)},$$

where γ is aggregate risk tolerance. In equilibrium, $\phi > 0$: rational momentum traders must trend-chase.

Model output: hump-shaped impulse response

- Newswatchers only: gradual adjustment to long-run value.
- Momentum traders: prices peak above fundamentals, then converge back through damped oscillations.
- Positive short-run correlation implies eventual negative autocorrelation under risk neutrality.



Comparative statics

- **Horizon j:** longer horizons generally delay the first negative autocorrelation; peak timing is tied to j.
- **Risk tolerance γ:** more aggressive momentum trading raises volatility and peak overshoot; pricing error is U-shaped.
- **Slower diffusion z:** larger momentum opportunities, stronger continuation, and more pronounced reversals.
- **Contrarian/smart-money arbitrage:** attenuates, but does not eliminate, the dynamics unless arbitrage capacity is unlimited.

Testable cross-sectional implications

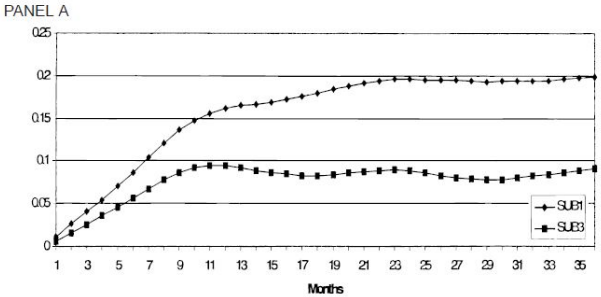
- Momentum is strongest among small, low-analyst-coverage stocks where information diffuses slowly.
- The same stocks should also be most reversal-prone at longer horizons.
- Public news may drift without the same overreaction if traders can condition on announcement dates.
- Higher trading costs / longer horizons keep return autocorrelations positive longer.

Takeaway

The model unifies momentum and reversal through *heterogeneous-agent interaction*, not a single representative-agent bias.

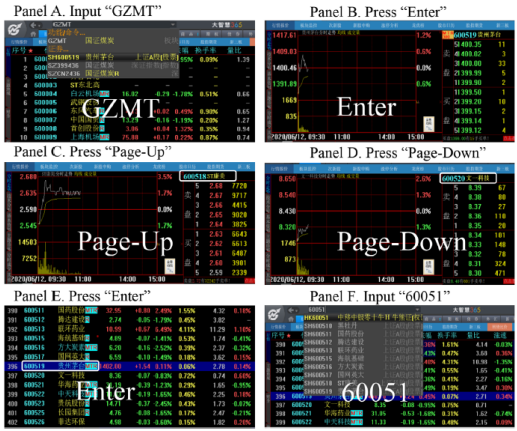
Limited Attention III: Empirical Test

- Hong, Lim, and Stein (2000)
- Test the Hong and Stein (1999) hypothesis that momentum is due to slow infusion of private information
  - They find that momentum is stronger among small stocks and stocks with low analyst coverage



Causal Effect of Attention on Asset Prices

- The key challenge in identifying the causal effect of attention on asset prices is that investor attention is endogenous to stock fundamentals (possibly unobservable)
- Chen, An, Wang, and Yu (JF 2023) circumvent this issue by exploring the attention spillover effect
- They exploit the unique setting in China where the on-screen display order of stocks is determined by the stock's listing code
- Investors are more likely to pay attention to neighboring stocks if they have positive investment experience with the focal stock



Chen, An, Wang, and Yu (JF 2023)

Causal Effect of Attention on Asset Prices, ctd.

- Because attention leads to excessive buying demand, stocks displayed next to those with higher returns in the past two weeks are associated with higher returns in the following week, which revert in the long run

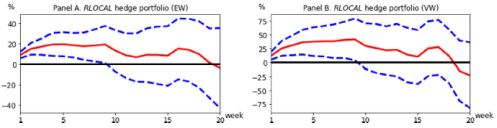


Figure 5. Cumulative return of the long-short portfolio based on *RLOCAL*. This figure shows the annualized cumulative CH4-alpha (in percentage points) to the equal- and value-weighted long-short portfolios (P5-P1) based on *RLOCAL* from week $t+1$ to week $t+20$, as well as the 95% confidence interval. For each stock, *LOCAL* is calculated as the value-weighted average return over the past two weeks of the 10 closest stocks on screen display, and *RLOCAL* is the residual of the cross-sectional regression of *LOCAL* on the focal stock's return over the past two weeks. Portfolios are formed at the end of week t based on *RLOCAL*. (Color figure can be viewed at wileyonlinelibrary.com)

Chen, An, Wang and Yu (JF 2023)

Category Thinking

Category Thinking

- A basic feature of the way humans think about a complex world is that we put things into categories and form beliefs at the level of these categories (Rosch and Lloyd, 1978)
- It is plausible that such category-based thinking also applies to investment
  - As there are thousands of individual securities and funds, investors may simplify their decision-making by putting financial assets into categories
  - We often use labels such as "value stocks," "growth stocks," "small-cap stocks"
- **Barberis and Shleifer (2003)**
  - Model where some investors put assets into categories and form beliefs about the assets' future returns at the category level
- A novel prediction is **excess return comovement**
  - Returns of assets in the same category comove more than can be explained by fundamentals, driven by correlated demand shocks

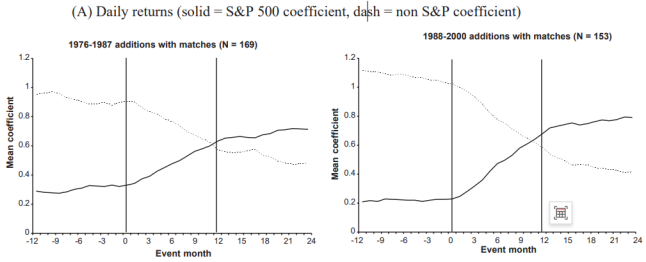
Category Thinking, ctd.

- Numerous studies document excess return comovement among firms or funds sharing similar characteristics, supporting Barberis and Shleifer (2003):
  - Stocks belonging to the same indices (Barberis et al., 2005; Greenwood, 2008; Boyer, 2011)
  - Firms headquartered in the same state (Pirinsky and Wang, 2006)
  - Stocks with similar levels of prices (Green and Hwang, 2009)
  - Firms covered by similar sets of analysts (Israelsen, 2016)
  - Stocks held by a common set of mutual funds (Anton and Polk, 2014)
  - Dividend-paying stocks (Hameed and Xie, 2019)

Excess Return Comovement among Stocks in the Same Index

- Barberis, Shleifer, and Wurgler (2005) test the prediction of category thinking that a stock that is added to (removed from) the S&P 500 index will experience a large increase (decrease) in its loading on the S&P return

$$R_{j,t} = \alpha_j + \beta_{j,\text{SP500}} R_{\text{SP500},t} + \beta_{j,\text{nonSP500}} R_{\text{nonSP500},t} + v_{j,t}$$



Psychology-free Approaches

Psychology-free Approaches

- The approaches discussed so far have mostly been rooted in a psychological concept
- Some behavioral finance models shed light on the asset pricing facts without appealing to any specific psychology
- A common structure for such a model is De Long et al. (1990), where less than fully rational investors ("noise traders") interact with more rational traders known as "arbitrageurs"
- Even without additional psychological assumptions, it yields many testable predictions:
  - The level of mispricing will be higher for assets whose valuation is more uncertain
  - The level of mispricing will be higher for assets whose characteristics make arbitrage more difficult
  - Arbitrageurs' actions in correcting mispricing are informative about future asset returns

Actions of Sophisticated Agents

- If some assets are mispriced, the trades of sophisticated investors should predict returns
- **Who are these sophisticated investors?**
- **Firm managers**
  - The key assumption is that firm managers can detect and respond to market mispricing
  - Issuance/repurchase decisions: change the supply of shares in response to investor demand
  - Personal transactions in their own firms' stocks — "corporate insider trading"
- **Short sellers**
  - Stocks heavily shorted have low subsequent returns (Asquith, Pathak, and Ritter, 2005)
- **Hedge funds**
  - Chen, Da, and Huang (2018): net arbitrage trading (NAT) of hedge funds strongly predicts stock returns in the cross-section
- **Analysts?**
  - Most studies show analyst recommendations and price targets are biased and overly optimistic
  - Lee and So (2017) show that abnormal analyst coverage positively predicts subsequent return
  - Look at what analysts do, not what they say

Actions of Sophisticated Agents, ctd.

- Interestingly, actions of sophisticated investors, when aggregated to the market level, can predict market returns in the same direction as stock-level returns
  - Baker and Wurgler (2000): the share of equity issues in total equity and debt issues negatively predicts market returns
  - Seyhun (1988): aggregate trading by corporate insiders positively predicts future market returns
  - Rapach, Ringgenberg, and Zhou (2016): aggregate short interest negatively forecasts market returns

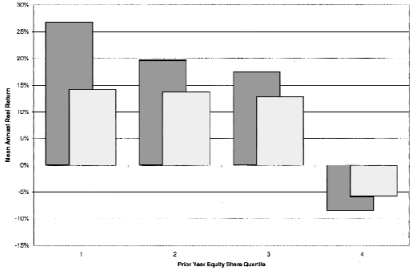


Figure 2. Mean equity returns by prior-year equity share in new issues, 1928–1997. Mean annual real returns on the CRSP value-weighted (light) and equal-weighted (solid) indexes by quartile of the prior-year share of equity issues in total equity and debt issues. Real returns are created using the Consumer Price Index from Ibbotson Associates (1998).

Baker and Wurgler (2000)

Actions of Noise Traders

- If some assets are mispriced, the trades and actions of unsophisticated agents should negatively predict returns
- **Several studies confirm this:**
- **Frazzini and Lamont (2008)**
  - Stocks held by mutual funds with large inflows in the previous three years have low subsequent returns
- **Barber, Odean, and Zhu (2008) and Hvidkjaer (2008)**
  - Stocks heavily bought by individual investors in the previous year perform poorly in the following year
  - Use small trades in the TAQ dataset to identify individuals and the Lee-Ready algorithm to identify buy/sell orders
- **Da, Engelberg, and Gao (2011)**
  - Higher investor attention as captured by Google search volume is correlated with abnormal retail buying and leads to elevated stock price initially but lower returns over the long run

My Research: Information Acquisition via EDGAR Search

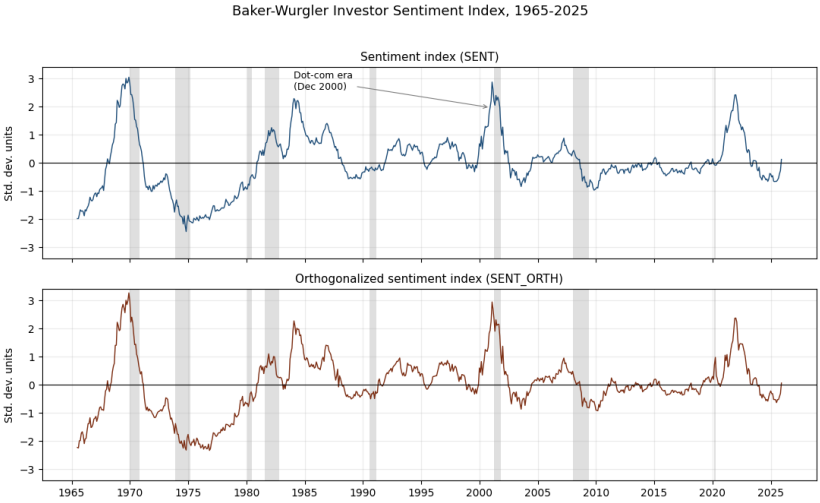
- **Li and Sun (JEDC 2022)**
  - SEC EDGAR server logs record the number of IP addresses searching each firm's filings, a direct measure of costly information acquisition
  - Abnormal EDGAR search predicts future returns and fundamentals
  - A long-short portfolio earns 80 bps monthly with no long-run reversal
  - Predictability is stronger for larger and lengthier filings that cost more to process, consistent with endogenous information acquisition
- **Connection:** Google search volume measures what grabs retail investors' attention; EDGAR search measures deliberate, costly processing by sophisticated investors. The two behave oppositely: attention-driven demand reverses, informed acquisition does not

Investor Sentiment

Market-based Measures of Sentiment

- Baker and Wurgler (2006)
- Measure investor "sentiment" using the principal component of the following six proxies from market data:
  - **Trading volume:** TURN is defined as the natural logarithm of the ratio of NYSE trading volume to the number of shares listed, detrended by the 5-year moving average
  - **Closed-end fund discount:** CEFD is defined as the average difference between the net asset values of closed-end stock fund shares and their market prices
  - **IPO activity:** NIPO is defined as the number of IPOs
  - **First-day returns:** RIPO is defined as the average of first-day returns on IPOs
  - **Equity issues:** S is defined as the ratio of gross equity issuance to the sum of gross equity and gross long-term debt issuance by all corporations
  - **Dividend premium:** DP is defined as the difference between the average market-to-book ratios of dividend payers and nonpayers
- Each of those measures has been proposed by prior studies as capturing investor sentiment

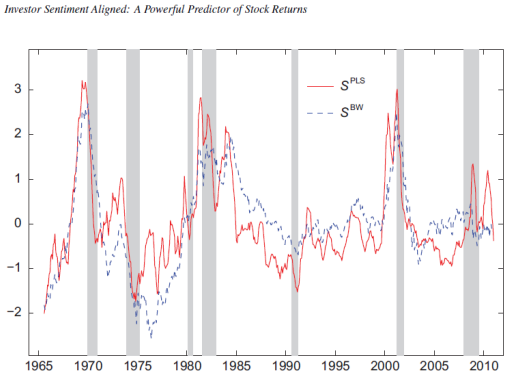
Baker and Wurgler Sentiment Index



Source: Baker and Wurgler (2006); Huang, Jiang, Tu, and Zhou (2015)

Baker and Wurgler Sentiment Index

- Huang, Jiang, Tu, and Zhou (2015)
- Refine the BW sentiment index by using the partial least squares (PLS) regression to more efficiently extract the return forecasting information from the six underlying proxies
- They show the new index has much greater predictive power than the original sentiment indices, both in and out of sample

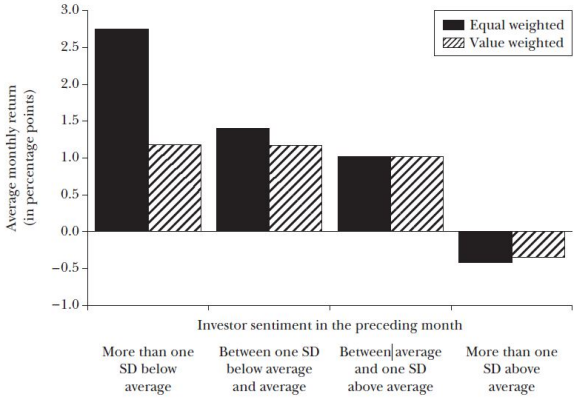


Huang, Jiang, Tu, and Zhou (2015)

Effects of Investor Sentiment on the Stock Market (I)

- Baker and Wurgler (2007): Sentiment should predict market returns with a negative sign

Figure 6
Sentiment and Market Returns



Effects of Investor Sentiment on the Stock Market (I) — Role of Uncertainty

- Birru and Young (JFE 2022) show that the return predictability of investor sentiment is much greater when uncertainty is high
- This is consistent with the idea that sentiment exhibits its strongest effects on asset prices at times when valuations are most subjective

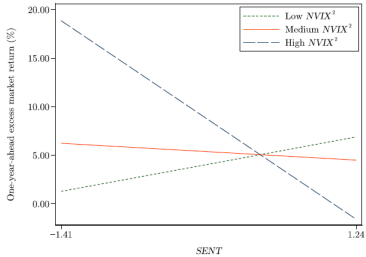


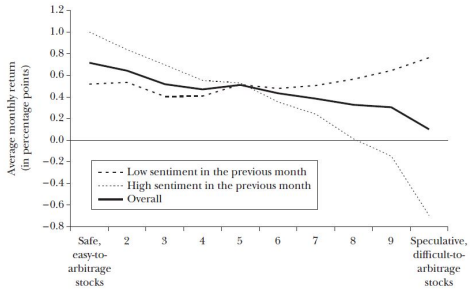
Fig. 1. Sentiment and Market Returns. *Panel A: Daily* Panels A and B plot the relation between sentiment and future market returns conditional on low, medium, and high uncertainty at the daily and monthly frequency, respectively. Equation (1) provides the parameter estimates. Low, medium, and high uncertainty correspond to the 10th, 50th, and 90th percentile of the distribution, respectively. In Panel A, *VIX* is the CBOE volatility index, the x-axis is the range between the 10th and 90th percentile of the Da et al. (2015) sentiment index (*FEARS*), and the y-axis is the one-day ahead excess market return. In Panel B, $NVIX^2$ is the square of the news implied volatility index from Manela and Moreira (2017), the x-axis is the range between the 10th and 90th percentile of the Baker and Wurgler (2006) sentiment index (*SENT*), and the y-axis is the one-year ahead excess market return. The sample period spans July 2004 to December 2011 for Panel A and July 1965 to March 2016 for Panel B.

Birru and Young (JFE 2022)

Effects of Investor Sentiment on the Stock Market (II) — Cross-Sectional Implication

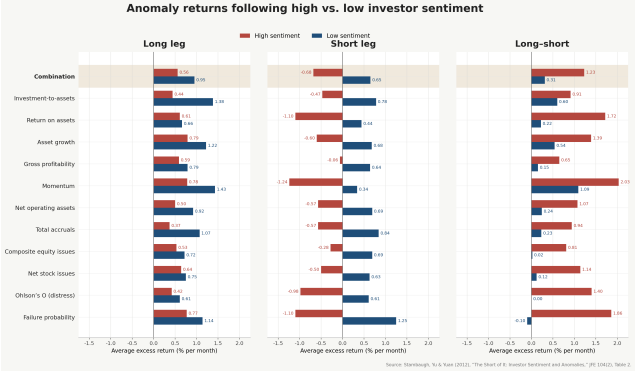
- When market-wide sentiment is high, stocks that are harder to arbitrage and more difficult to value will be more overpriced and have lower subsequent returns
- Baker and Wurgler (2006) confirm this prediction
  - Identify hard-to-arbitrage and difficult-to-value stocks as small stocks, young firms, volatile stocks, unprofitable stocks, and non-dividend-paying stocks

Figure 5
Sentiment and Future Returns



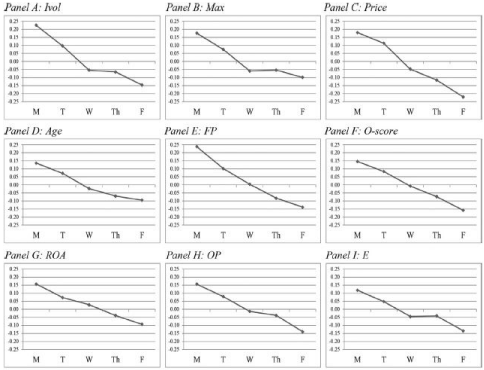
Effects of Investor Sentiment on the Stock Market (III) — Anomalies

- Stambaugh, Yu, and Yuan (JFE 2012) argue that anomalies should be amplified following high sentiment
  - The effect should concentrate on the short leg due to short-sale constraints



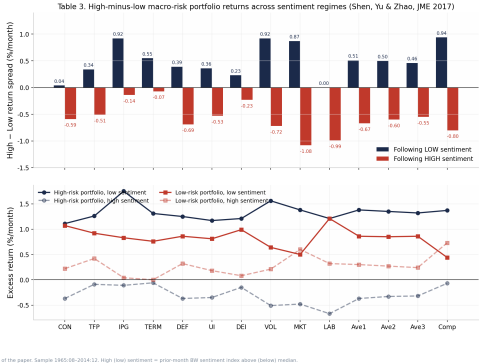
Effects of Investor Sentiment on the Stock Market (III) — Anomalies, ctd.

- Birru (JFE 2018) documents that long-short anomaly returns are strongly related to the day of the week
  - In particular, long-short strategies earn the highest return on Monday and the lowest return on Friday; the effect concentrates in the short leg, which is more speculative
  - Consistent with the psychology literature that mood increases (decreases) on Friday (Monday)



Effects of Investor Sentiment on the Stock Market (IV) — Risk-Return Trade-off

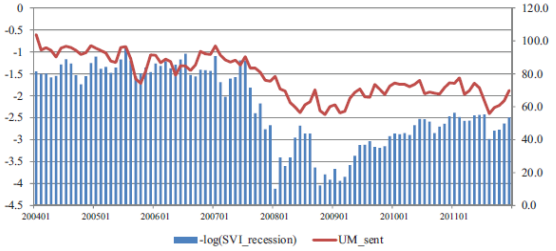
- Shen, Yu, and Zhao (JME 2017) argue that sentiment also affects the cross-sectional risk-return trade-off
  - When sentiment is high, sentiment-driven investors tend to require a smaller compensation for the risk they bear
  - In addition, riskier stocks may be more subject to sentiment shifts and become more overpriced
  - Coupled with short-sale constraints, the risk-return trade-off may be attenuated or even reversed during high sentiment periods



Data: Table 3 of the paper. Sample 1965:08–2014:12. High (low) sentiment = prior-month BW sentiment index above (below) median.

Other Measures of Aggregate Sentiment — Google Search

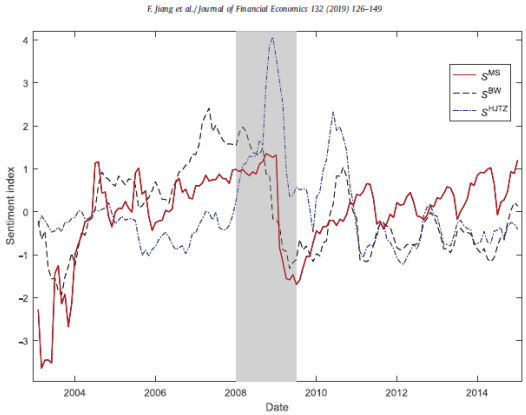
- Da, Engelberg, and Gao (RFS 2015)
- Aggregate the volume of Google search queries related to household concerns and construct a Financial and Economic Attitudes Revealed by Search (FEARS) index as a new measure of investor sentiment
- Benefits of FEARS:
  - More transparent
  - Higher frequency — daily
  - Reveal attitudes rather than inquire about them
- Gao, Ren, and Zhang (2020) document similar evidence for international markets



- FEARS is negatively related to contemporaneous returns
- FEARS is positively related to future returns
- FEARS predicts a temporary increase in volatility
- FEARS predicts mutual fund flows out of equity funds to bond funds

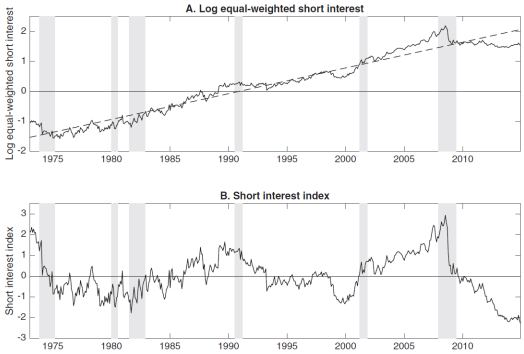
Other Measures of Aggregate Sentiment — Manager Sentiment

- Jiang et al. (JFE 2019)
  - Manager sentiment index based on the aggregated textual tone in 10-Ks, 10-Qs, and conference call transcripts
  - Manager sentiment is a strong negative predictor of future aggregate stock market returns



Other Measures of Aggregate Sentiment — Aggregate Short Interests

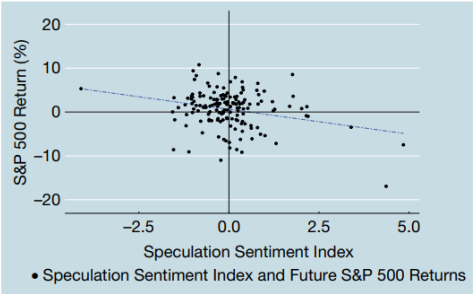
- Short sellers are sophisticated investors who can correct mispricing induced by sentiment traders
- Short interests aggregated to the market level are a strong contrarian predictor of future market returns



Other Measures of Aggregate Sentiment — Flows to Leveraged ETFs

- Leveraged ETFs provide amplified gains/losses to investors who bet on the direction of future market movement
- Heavily traded and cater to speculative investors
- Hence flows to such ETFs mainly reflect speculative investors' demand
- Davies (2022) constructs a Speculative Sentiment Index (SSI) based on this insight
  - SSI is the difference between share changes in the leveraged-long ETFs and share changes in the leveraged-short ETFs
- A 1-standard-deviation increase in the monthly index is associated with a 1.14%–1.67% decline in broad-market stock returns in the next month

Graph C. SSI Versus S&P 500 r_{t+1}



Other Measures of Aggregate Sentiment — LLM-Generated Beliefs (I)

Bybee (U. Chicago Booth), Feb. 2025, "The Ghost in the Machine" | Can LLMs serve as *measurable proxies for human beliefs*?

1. Method: beliefs from text

- Beliefs are encoded in text ⇒ LLMs trained on massive corpora absorb the same world models *and biases*
- Input: 300 WSJ articles/month (1984–2021) → GPT-3.5 prompted for {*increase/decrease*, confidence, magnitude, explanation} on 14 macro/finance series
- Aggregate to a *balance statistic*: confidence-weighted $\big(\#\text{"increase"} - \#\text{"decrease"}\big)\big/\big(\#\text{"increase"} + \#\text{"decrease"}\big)$ for each series
- BERT + ridge *transfer learning* infills full corpus; NYT extends sample back to 1900

2. Validation: GPT ≈ human surveys

- Correlates with AAII & CFO surveys (avg. 0.47 vs. 0.43 between surveys); matches SPF revisions in 11/13 series
- Reproduces deviations from FIRE: extrapolative (+ past returns), + fund flows, − objective ER proxies (D/P, CAY), − future returns
- Mistakes traceable: GPT expects high future returns because of high future *earnings* — neglect of general equilibrium (cf. Andre et al. 2024)
- Out-of-sample (post-training news): correlations persist ⇒ not look-ahead bias

Other Measures of Aggregate Sentiment — LLM-Generated Beliefs (II)

3. 120 years of economic sentiment

- Factor model of GPT expectations: extract sentiment δ_t as the first principal component, orthogonalized w.r.t. the rational forecast (ex-ante dynamic factor / ex-post realizations)
- Spikes at sensible events: 1929, Lehman, Covid ($-$); dot-com, FDR 1936 ($+$); corr. 0.54 with Shiller CAPE
- Responds $+$ to past returns; $-$ predicts future returns (-0.41 @ 24m, recent sample)

4. Application: behavioral bubbles

- Mispricing proxy: rolling sentiment betas (sensitivity of industry dividend growth to δ_t) on Greenwood et al. (2019) 40 industry run-ups
- Run-ups that *crash* have higher sentiment betas $\Rightarrow$ higher crash prob., lower future returns (robust to controls)
- Stronger return $\rightarrow$ sentiment feedback during crashing run-ups $\Rightarrow$ extrapolation as a key mechanism
- Threshold strategy avoids $\sim$80% of crashes, earns significant abnormal returns

Takeaway: LLMs are not super-forecasters — they generate *human-like beliefs, biases included.* This gives us measurable expectations wherever news exists, at daily frequency, over a century+ — a new tool for studying extrapolation, sentiment, and bubbles.

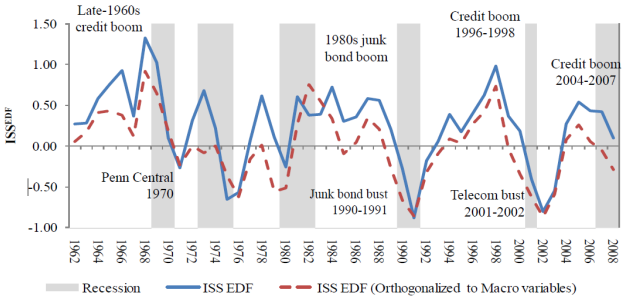
Measuring Investor Mood/Emotions

- Edmans et al. (2022): Music sentiment
- The positivity of songs that individuals choose to listen to
- **Advantages:**
  - Real-time, continuous measure of national sentiment
  - Language-free and comparable globally
  - Exogenous to financial market events
- **Main findings:**
  - Music sentiment is positively correlated with same-week equity market returns and negatively correlated with next-week returns
  - Music sentiment also predicts increases in net mutual fund flows and precedes a rise in stock market volatility
  - Negatively associated with government bond returns, consistent with a flight to safety

Credit Market Sentiment

Credit Market Sentiment

- Greenwood and Hanson (RFS 2013) construct a sentiment measure for the credit market
  - The idea is to compare the credit quality of high-issuance firms with that of low-issuance firms
  - The insight is that a significant decline in issuers' credit quality is a reliable signal of credit market overheating



Credit Market Sentiment Forecasts Excess Corporate Bond Returns

- Economic magnitudes are significant:
  - A 1-σ increase in ISS EDF, the issuer expected default frequency, $\Longrightarrow$ cumulative excess returns fall by 7.30% over the following 2 years

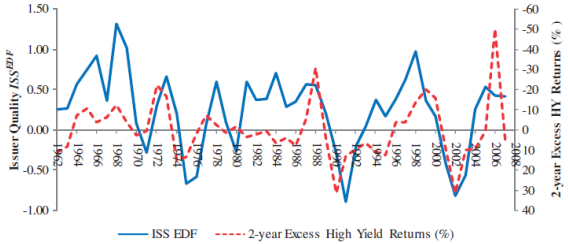


Figure 3
Issuer quality and subsequent high-yield excess returns
Issuer quality (left axis) plotted alongside cumulative excess high-yield bond returns for the following two years (right axis). Returns are plotted in reverse scale, so the negative correlation appears positive visually. Issuer quality is measured with ISS^{EDF}, the difference between the average *EDF* decile of high- and low-debt issuers from 1962–2008.

Greenwood and Hanson (RFS 2013)

Credit Market Sentiment and Economic Growth

- Credit market overheating is associated with a higher probability of a future financial crisis and of economic downturns
  - Schularick and Taylor (2012); Lopez-Salido, Stein, and Zakrajsek (2017); Baron and Xiong (2017)

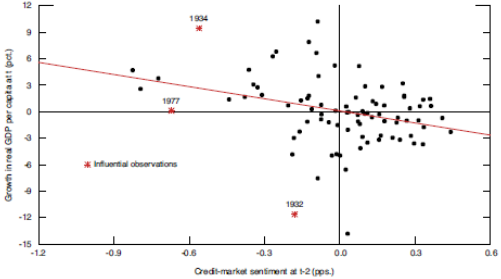


FIGURE II

Credit-Market Sentiment and Economic Growth, 1929–2015

The scatter plot depicts a visual representation of the relationship between credit-market sentiment in year $t-2$ and the growth in real GDP per capita from year $t-1$ to year t, implied by the specification in column (1) of Table II. See the text and Figure III for the definition of influential observations

Lopez-Salido, Stein, and Zakrajsek (QJE 2017)

Credit Expansion Predicts Banking Crises

- Baron and Xiong (QJE 2017) find bank credit expansion predicts increased bank equity crash risk
- Despite the elevated crash risk, credit expansion predicts lower mean bank equity returns in the subsequent one to three years
- Bank credit expansion is distinct from equity market sentiment captured by dividend yield
- The result suggests overoptimism and neglect of crash risk by bank equity investors during credit expansions

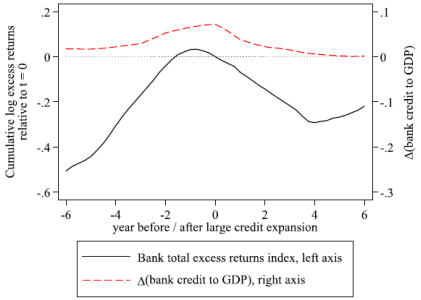


FIGURE II

Bank Equity Prices and Bank Credit before and after Large Credit Expansions

The past three-year change in bank credit to GDP (i.e., $\Delta(\frac{bank\ credit}{GDP})$) and the bank total excess log returns index are plotted before and after a large credit expansion. A large credit expansion is defined as *credit expansion* exceeding the 95th percentile threshold, which is calculated for each country and each point in time using only past information to avoid any future-looking bias. $\Delta(\frac{bank\ credit}{GDP})$ and bank total excess log returns are pooled averages across time and countries, conditional on the given number of years before or after the start of a banking crisis. The average bank log returns are then cumulated from $t = -6$ to $t = +6$, and the level is adjusted to be 0 at $t = 0$. Observations are over the sample of 20 countries, 1920–2012.

Baron and Xiong (QJE 2017)

Predictable Financial Crisis

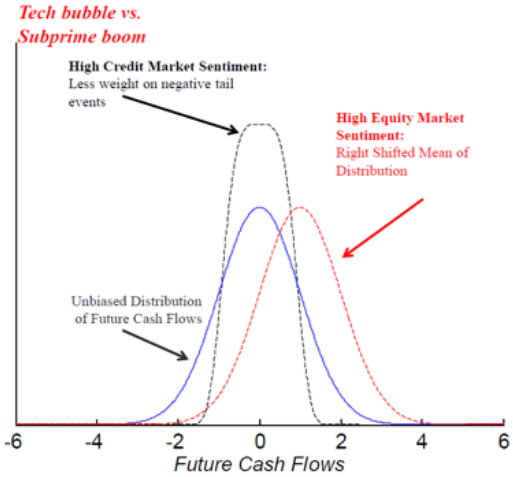
- Greenwood, Hanson, Shleifer, and Sorensen (JF 2022)
- They use historical data on postwar financial crises around the world
- The combination of rapid credit and asset price growth over the prior three years (R-zone) is associated with a 40% probability of entering a financial crisis within the next three years

Panel A: Business Sector

	Crisis within One Year				Crisis within Two Years				Crisis within Three Years				Crisis within Four Years			
	(1.1)	(1.2)	(1.3)	(1.4)	(2.1)	(2.2)	(2.3)	(2.4)	(3.1)	(3.2)	(3.3)	(3.4)	(4.1)	(4.2)	(4.3)	(4.4)
High Debt Growth$^{Bus.}$ (β^h)	6.9**		5.3**		11.6***		9.5**		16.8***		11.5**		15.6**		10.3*	
	[2.3]		[2.1]		[3.0]		[2.5]		[3.3]		[2.7]		[2.7]		[2.2]	
High Price Growth$^{Bus.}$ (δ^h)		0.4	−0.4			4.8	3.8			10.5	7.4			10.7	7.6	
		[0.1]	[−0.2]			[0.9]	[0.8]			[1.4]	[1.1]			[1.5]	[1.2]	
R-Zone$^{Bus.}$ (γ^h)			5.3	9.0			7.8	17.9*			19.4**	33.7***			19.4**	33.0**
			[0.8]	[1.1]			[1.3]	[2.1]			[2.8]	[3.3]			[2.6]	[3.1]
Sum of coefficients ($\beta^h + \delta^h + \gamma^h$)	6.9	0.4	10.2	9.0	11.6	4.8	21.1	17.9	16.8	10.5	38.2	33.7	15.6	10.7	37.3	33.0
t-Statistic ($\beta^h + \delta^h + \gamma^h$)			1.2				2.1				3.2				3.1	
R^2 *(within)*	1.6	0.0	1.9	1.1	2.5	0.7	3.6	2.3	3.8	2.4	7.8	6.1	2.8	2.1	6.2	4.8
N	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258	1258

Credit Market vs. Equity Market Sentiment

- Empirical measures of credit market sentiment are only modestly correlated with measures of equity market sentiment
  - Suggests we should not think of equity and credit markets as fully integrated
- Measures of expected returns on the stock market, such as Shiller's CAPE, do not predict future GDP growth (Lopez-Salido et al., 2017)
  - Suggests credit market sentiment has potentially greater macroeconomic significance



Narratives of the Credit Cycle

- Short-term rates are low or have fallen in the recent past
  - Push investors to "reach for yield" by buying bonds with longer duration and lower credit quality
- Term and credit spreads become compressed as a result
- Default rates have been low or falling, because lower credit spreads allow low-quality firms to obtain cheap financing
- This credit boom incentivizes low-quality firms to take on excessive leverage, making them vulnerable to even small external shocks
- Consistent with empirical evidence on the boom-bust credit cycle

How Credit Cycles across a Financial Crisis

- Krishnamurthy and Muir (JF 2025)
- Analyze the behavior of credit and output in financial crises using data on credit spreads and credit growth
- Precrisis spreads tend to drop to low levels while credit growth accelerates

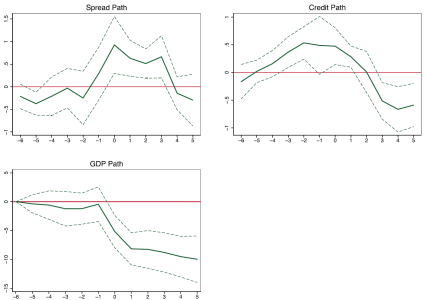


Figure 1. Behavior of spreads, credit, and GDP. This figure plots the behavior of credit spreads, GDP, and the quantity of credit around crises, with date 0 as the JST mod. date associated with the 40 JST crises. GDP is expressed as the percentage fall from year $t-6$, after demeaning by the growth rate in GDP for country-i. Credit is the three-year growth in credit-to-GDP, demeaned by the country-i growth rate and normalized by the standard deviation of credit growth for the country. Spreads are normalized by dividing by the unconditional mean consistent with our normalization in the main text and plotted as deviations from the country-i mean. See the text for further details. The dashed lines indicate 95% confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

Early Indicator of the Credit Boom

- Ben-Rephael, Choi, and Goldstein (JFE 2021)
- Early indicator of credit boom based on intra-family flow shifts toward high-yield (HY) bond mutual funds
- Intra-family flows likely capture investors' deliberate asset allocation decisions (rather than saving/investment decision)
- It leads other indicators of the credit boom, such as growth in financial intermediary balance sheets, increases in shares of high-yield bond issuers, and downturns in various measures of credit spreads
- Attributes the origins of the credit boom to an increase in investors' demand for HY bonds, perhaps due to biased perceptions of default risk

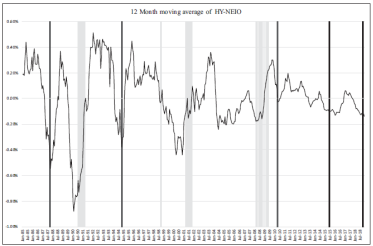


Fig. 1. 12-month moving averages of HY-NEIO. In this figure we plot the 12-month moving averages of HY-NEIO from January 1985 (i.e., February 84–January 85) through December 2018. HY-NEIO is net exchanges (exchanges-in minus exchanges-out) from high-yield corporate bond funds normalized by the end-of-previous-month assets. The data are obtained from the 83 and out from February 1984 through December 2018. The three light gray columns represent NBER recession periods. The dark gray columns represent the 1987 market crash, the Mexican Peso crisis in 1994, the European sovereign debt crisis in early 2010, the high-yield spread hike during June 2015–February 2016, and the market crash in the last quarter of 2018.

A Behavioral Account of the Credit Cycle

- Narrative of the credit cycle from Minsky's financial instability hypothesis and Kindleberger's *Manias, Panics, and Crashes: A History of Financial Crises*
- The full model is developed in Bordalo et al. (2018) and Greenwood, Hanson, and Jin (2024)
- Key feature of the model is **reflexivity**: the feedback loop between investors' biased perceptions and market outcomes
- **Main idea:**
  - Investors hold extrapolative beliefs based on borrowers' recent default history
  - During a boom, investors extend credit on attractive terms because borrowers have paid back in the past
  - Even if borrowers' fundamental positions deteriorate, this may not be directly visible for a while because borrowers roll over debt at low rates
  - When the reckoning eventually comes, markets overreact, slowing the recovery

Summary

- **Bounded rationality:**
  - Limited attention leads to underreaction to news that is less salient or more costly to process
  - Category thinking leads to excess return comovement
- **Psychology-free approaches:**
  - Actions of sophisticated traders predict returns in the right direction
  - Actions of noise traders negatively predict returns
  - Investor sentiment predicts market returns and cross-sectional anomalies
- **Credit market sentiment:**
  - Distinct from equity market sentiment
  - Predicts financial crises and economic downturns
  - Behavioral models explain credit cycle dynamics